

SWAYAM Prompt Engineering - Week 2 Short Notes

1. Unlocking AI Precision: Three-Step Response Elevation Framework

Generate an initial AI response, evaluate it for accuracy, clarity and completeness, then refine the prompt with more context, audience and format. Example: Instead of 'Explain AI', ask 'Explain AI for first-year engineering students with examples in 300 words.'

2. Text Mastery – Summaries, Extraction & Translation

AI can summarize long documents, extract key details such as names, dates and findings, and translate text while preserving meaning. Example: Summarize a research paper, extract objectives and results, then translate the summary into Telugu.

3. Text Mastery – Summarisation & Structured Extraction

Summarisation reduces lengthy text into concise points. Structured extraction organizes information into tables, bullet points or JSON. Example: Extract a resume into Name, Skills, Education, Projects and Experience.

4. NotebookLM Audio Overview

NotebookLM converts uploaded documents into podcast-style conversations based only on the uploaded sources. It helps learners revise while listening. Example: Create a 10-minute podcast explaining a chapter on Machine Learning.

5. Seeing Knowledge: Video Overview, Mind Maps & Infographics

NotebookLM creates visual learning resources such as mind maps, infographics and video summaries. Example: Upload an AI chapter and generate a mind map connecting Machine Learning, Deep Learning and NLP.

6. NotebookLM as a Research Assistant

It summarizes research papers, compares multiple papers, prepares literature reviews and identifies research gaps. Example: Compare three blockchain papers by objective, dataset, methodology and results.

7. From Output to Insight: Evaluating, Validating & Refining AI Responses

Always evaluate AI outputs for accuracy, relevance and completeness. Validate facts using trusted sources and refine prompts for better responses. Example: Ask AI to review and improve its previous answer after fact-checking.

8. Spreadsheet to Story: Data Analysis with Julius.ai

Julius.ai analyzes Excel or CSV files, creates charts, dashboards and presentations, and explains trends in simple language. Example: Upload monthly sales data to identify

growth trends and generate a presentation.